平行于外磁场传播的低频磁流体波.

$\hat{k} = k\hat{z}$ $\quad B = B_0\hat{z}$ $\quad \vec{E}_1 = E_1\hat{x}$ $\quad B_1 = B_1\hat{y}$.

$$\nabla \times \vec{E} = -\frac{\partial}{\partial t}\vec{B}$$

$$\nabla \times \vec{B} = \frac{1}{c^2}\frac{\partial \vec{E}}{\partial t} + \mu_0 \vec{j}$$

$$\nabla \times (\nabla \times \vec{E}) = -\frac{\partial}{\partial t}(\nabla \times \vec{B})$$

$$\nabla \times (\nabla \times \vec{E}_1) = -\vec{k} \times (\vec{k} \times \vec{E}_1)$$

由 $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a}\cdot\vec{c}) - \vec{c}(\vec{a}\cdot\vec{b})$

$$\nabla \times (\nabla \times \vec{E}_1) = k^2 \vec{E}_1$$

$$-\frac{\partial}{\partial t}(\nabla \times \vec{B}) = -\frac{\partial}{\partial t}\left(\frac{1}{c^2}\frac{\partial}{\partial t}\vec{E}_1 + \mu_0 \vec{j}\right)$$

$$= -\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\vec{E}_1 - \mu_0 \frac{\partial}{\partial t}\vec{j}$$

$$k^2 \vec{E}_1 = \frac{\omega^2}{c^2}\vec{E}_1 + i\omega\mu_0 \vec{j}$$

其中, $\vec{j}_1 = n_0 e(\vec{v}_{ix} - \vec{v}_{ex})$

运动方程：

$$M\frac{d}{dt}\vec{v}_i = e(\vec{E}_1 + \vec{v}_i \times \vec{B})$$

分量形式：

$$\begin{cases} -i\omega M v_{ix} = eE_{1x} + ev_{iy}B_0 \\ -i\omega M v_{iy} = -ev_{ix}B_0 \end{cases}$$

$$v_{iy} = -\frac{ieB_0}{\omega M} v_{ix}$$

$$-i\omega M v_{ix} = eE_1 - \frac{ie^2B_0^2}{\omega M} v_{ix}$$

$$\Rightarrow \quad v_{ix} = \frac{-ieE_1\omega M}{e^2B_0^2 - \omega^2M^2} = \frac{ieE_1}{\omega M}\left(\frac{\omega^2M^2}{\omega^2M^2 - e^2B_0^2}\right)$$

$$= \frac{ieE_1}{\omega M}\left(1 - \frac{e^2B_0^2}{\omega^2M^2}\right)^{-1}$$

$\frac{e^2B_0^2}{\omega^2M^2} = \Omega_c^2/\omega^2$ $\quad \Omega_c = \frac{eB_0}{M}$

$$= \frac{ieE_1}{\omega M}\left(1 - \frac{\Omega_c^2}{\omega^2}\right)^{-1}$$

$$v_{iy} = -\frac{ieB_0}{\omega M}\frac{ieE_1}{\omega M}\left(1 - \frac{\Omega_c^2}{\omega^2}\right)^{-1} = \frac{e\Omega_c}{\omega^2 M}\left(1 - \frac{\Omega_c^2}{\omega^2}\right)^{-1}E_1$$

对电子，相当于作替换：

$$M \to m,\quad e \to -e,\quad \Omega_c \to -\omega_c.$$

$$\begin{cases} v_{ex} = \dfrac{-ie}{m\omega}\left(1-\dfrac{\omega_c^2}{\omega^2}\right)^{-1} E_1 \\ v_{ey} = \dfrac{e\omega_c}{m\omega^2}\left(1-\dfrac{\omega_c^2}{\omega^2}\right)^{-1} E_1 \end{cases}$$

考虑：$\omega^2 \ll \omega_c^2$.

$$\begin{cases} v_{ex} = \dfrac{-ie}{m\omega}\left(1-\dfrac{\omega_c^2}{\omega^2}\right)^{-1} E_1 \longrightarrow 0 \\ v_{ey} = -\dfrac{e\omega_c\omega^2}{m\omega^2\omega_c^2} E_1 = -\dfrac{eE_1}{m\omega_c} = \dfrac{E_1}{B_0} \end{cases} \qquad \omega_c = \frac{eB_0}{m}.$$

代入 Maxwell eqs：

$$k^2 E_1 = \frac{\omega^2}{c^2} E_1 + i\omega\mu_0 n_0 e \frac{ie}{M\omega}\left(1-\frac{\Omega_c^2}{\omega^2}\right)^{-1} E_1$$

$$(\omega^2 - c^2k^2) E_1 = \frac{\omega\mu_0 n_0 e^2 c^2}{M\omega}\left(1-\frac{\Omega_c^2}{\omega^2}\right)^{-1} E_1 \qquad \mu_0 c^2 = \frac{1}{\varepsilon_0}.$$

$$= \frac{n_0 e^2}{\varepsilon_0 M}\left(1-\frac{\Omega_c^2}{\omega^2}\right)^{-1} E_1 \qquad \Omega_p^2 = \frac{n_0 e^2}{\varepsilon_0 M}.$$

$$= \Omega_p^2\left(1-\frac{\Omega_c^2}{\omega^2}\right) E_1$$

约去 E_1.

$$\omega^2 - c^2k^2 = \Omega_p^2\left(1-\frac{\Omega_c^2}{\omega^2}\right)^{-1}.$$

考虑 $\omega^2 \ll \Omega_c^2$, $\Omega_c \sim 10^{-8}$

$$\omega^2 - c^2k^2 = -\omega^2 \frac{\Omega_p^2}{\Omega_c^2} = -\omega^2 \frac{e^2 n_0}{M\varepsilon_0} \cdot \frac{M^2}{e^2 B_0^2} = -\omega^2 \frac{n_0 M}{\varepsilon_0 B_0^2}$$

$$= -\omega^2 \frac{\rho}{\varepsilon_0 B^2} \quad , \quad (\rho \equiv n_0 M) .$$

$$\frac{\omega^2}{k^2} = v_p^2 = c^2 \left(1 + \frac{\rho}{\varepsilon_0 B^2}\right)^{-1} = \frac{c^2}{\varepsilon} \quad , \quad \varepsilon = \left(1 + \frac{\rho}{\varepsilon_0 B^2}\right) .$$

$$\frac{\omega}{k} = v_p = \frac{c}{\sqrt{\varepsilon}} .$$

考虑 $\varepsilon \gg 1$, $\quad v_p = \sqrt{\frac{c^2}{\rho/(\varepsilon_0 B_0^2)}} = \frac{cB_0}{\left(\frac{\rho}{\varepsilon_0}\right)^{1/2}} = v_A .$

$$\varepsilon = 1 + \frac{c^2}{v_A^2} .$$

Maxwell eqs :

$$\nabla \times \vec{E}_1 = -\frac{\partial}{\partial t} \vec{B}_1 \quad , \quad -i\vec{k} \times \vec{E} = i\omega \vec{B}_1$$

$$E_1 = \frac{\omega}{k} B_1 = v_A B_1$$